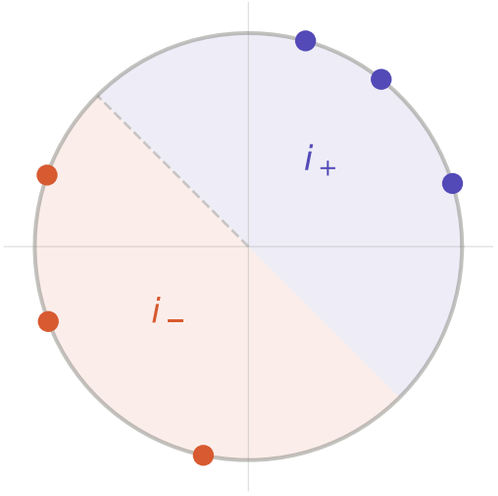
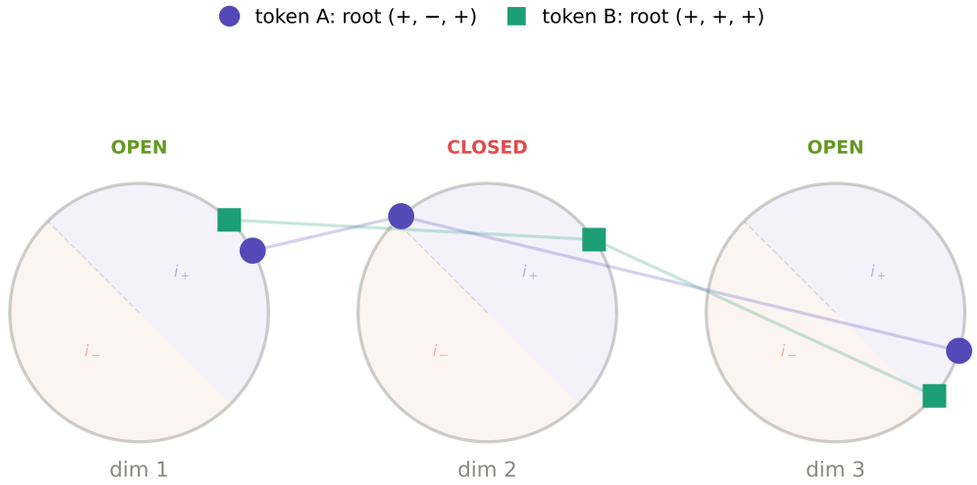


(a) $d = 1$: circle S^1, $2^1 = 2$ halves



(b) $d = 3$: two tokens on $\mathbb{T}^3$, root parity gates shown

score(A, B) per dimension

dim 1	roots: (+, +) → gate = 2 score = 2 × (1 + cos Δθ) = 3.91
dim 2	roots: (−, +) → gate = 0 score = 0 × anything = 0
dim 3	roots: (+, +) → gate = 2 score = 2 × (1 + cos Δθ) = 3.84

total score = 7.75 / 12

(blocked dims contribute exactly 0)

(c) root parity gate: per-dimension score